Real Estate App Development & Recommendation

FYP Progress Report

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Outline

Recap: Winter Break Progress

Completed:

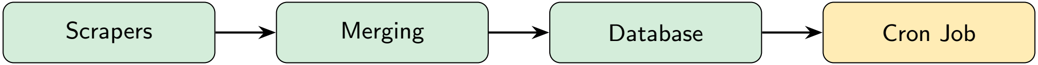
- Reliability Verification
- Parallelism Troubleshooting
- Operational Pivot to single-thread/proxy
- Pipeline Development (initial design)

Planned Future Work:

- Test Cron Job's Reliability
- Design Merging Logic (multi-site)
- Design Merging Logic (Gov data)
- Backend Development

Overview: Tasks Completed

Task	**Status**	**Completion**
Multi-Platform Scrapers	Operational	90%
Data Merging Pipeline	Operational	90%
Database Schema Design	Complete	100%
Cron Job Setup	Configured	90%



1. Multi-Platform Scrapers

Coverage: 4 Major Singapore Property Portals

PropertyGuru

- Tech: Requests + Proxy API
- Data: Sale & Rental listings
- Feature: Incremental update, auto-cleanup

99.co

- Tech: Playwright (headless)
- Data: Sale & Rental listings
- Feature: Retry logic, pagination

EdgeProp

- Tech: Playwright (headless)
- Data: HDB / Condo / Landed
- Feature: Debug HTML snapshots

SRX

- Tech: Playwright (headless)
- Data: Sale & Rental (by town)
- Feature: Concurrent crawling, CSV backup

2. Data Merging Pipeline (1/2)

Implementation: `pipeline/aggregate.py`

Standardization Functions:

- `normalize_propertyguru()`
- `normalize_99co()`
- `normalize_edgeprop()`
- `normalize_srx()`

Data Cleansing:

- `clean_price()` → numeric
- `clean_sqft()` → numeric
- `clean_tenure()` → unified
- `clean_property_type()`

2. Data Merging Pipeline (2/2)

Deduplication Logic:

- Composite Key:
  - Normalized Address
  - Beds + Baths + Sqft
- Source Priority: PropertyGuru first

Price Format Unification:

- Sale: `S$ 1,234,567`
- Rental: `S$ 3,500 /mo`
- PSF: `S$ 1,234.56 psf`

Output

Unified `aggregated_listings.csv` with consistent schema across all platforms

3. Database Architecture

Dual Database Support:

- PostgreSQL (production)
- SQLite (local dev)

Schema Design:

- `agents` table
- `listings` table
- FK: `agent_id`
- Unique: `url`

Key Fields in `listings`:

- title, address
- price (numeric), display_price
- psf (numeric), display_psf
- beds, baths, sqft
- property_type, tenure
- source, buy_rent
- created_at, updated_at

4. Cron Job Configuration

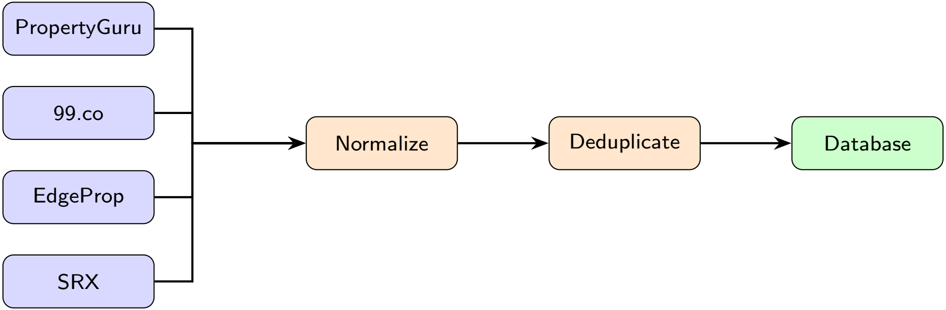
Automated Scheduling (Ready for Deployment):

Task	Schedule	Script
Daily Incremental Update	2:00 AM daily	`run_daily.py`
Weekly Cleanup	3:00 AM Monday	`run_cleanup.py`
Retry Failed Records	On-demand	`run_retry.py`

Crontab Example:

```
0 2 * * * python run_daily.py >> daily.log 2>&1
0 3 * * 1 python run_cleanup.py >> cleanup.log 2>&1
```

Data Flow Architecture



- **Input**: 4 platforms $\times$ 2 categories (Sale/Rent)
- **Output**: Unified CSV + PostgreSQL/SQLite database

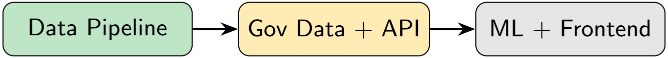
Future Work

Short-term (Next 2 Weeks):

1. Integrate Government Data
   - HDB info from data.gov.sg
   - URA transaction records
2. Backend API Development
   - FastAPI REST API
   - Search & Filter endpoints

Medium-term:

3. Recommendation System
   - Multi-factor analysis
   - Price prediction model
4. Frontend Development
   - Web/Mobile interface



Summary

Key Achievements (Week 3–4)

- Developed scrapers for 4 major Singapore property platforms
- Implemented data merging pipeline with standardization & deduplication
- Designed dual-database schema (PostgreSQL + SQLite)
- Configured automated cron job scheduling for daily updates

Next Priorities (Week 5–6)

- Integrate government data sources (HDB, URA via data.gov.sg)
- Develop FastAPI backend with RESTful endpoints

Thank You!

Questions?